

An Analysis of High RUL Prediction Challenges on the C-MAPSS FD001 Dataset: A Data-Centric Perspective

Erik Sun,¹

Abstract

The accurate prediction of Remaining Useful Life (RUL) for critical assets like aircraft engines is a major goal in predictive maintenance. The NASA C-MAPSS FD001 dataset is a widely used benchmark for developing and testing RUL prediction models. However, a common challenge is the difficulty in accurately predicting high RUL values. This article contends that this difficulty is not primarily due to the limitations of any specific model, but rather stems from the inherent characteristics of the dataset itself. Research has shown that a variety of models, including tree-based ensembles and deep neural networks, all struggle with high RUL predictions on this dataset. The core issues are the lack of clear degradation signals in the early life of the engines and the labelling scheme used in the dataset.

Keywords: keyword, keyword, keyword.

1 Lack of Degradation Trends in Early Life

In the FD001 subset, engines start in a near healthy state and exhibit minimal performance change until late in their life. During these early life windows (high RUL), there is an absence of clear degradation signal, which leads to model confusion. A recent study using a Bi-LSTM on the NASA turbofan model explicitly noted that:

accuracy during early life stages remains limited due to the absence of useful degradation signals¹

With no strong trend or feature variation to correlate with RUL in this phase, models essentially have to guess the RUL. This affects both classical regressors and deep learning models alike. For example, even an advanced deep network struggled in the early cycles until additional feature engineering was introduced². In practice, models often learn to predict a roughly constant high RUL in early cycles, since that fits the weak signal but this can be unstable and inaccurate for true RUL values far above the training mean. The literature consistently highlights that without explicit indicators of wear, the high-RUL predictions are unreliable and dominated by noise or initial condition variance, rather than meaningful degradation patterns¹.

2 Label Ambiguity and Identifiability Problems

A major source of difficulty is the piecewise RUL labelling strategy used in most C-MAPSS studies. It is standard to cap the RUL label at a fixed "maximum" value in the healthy phase (typically 125 cycles for FD001³). In other words, all data points where the true RUL > 125 are assigned the same label (125). This creates a long flat region in the target function:

RUL is initially allocated to a fixed number (known as early RUL)... When the RUL value hits this threshold, it then follows a linear decline⁴

¹540703757, esun0349@uni.sydney.edu.au

The consequences is severe label ambiguity in early life. Many engines have identical RUL labels (the max value) for a large number of initial cycles, even if some of those engines will ultimately run much longer than others. Thus sample with very similar sensor readings can correspond to significantly different actual lifetimes, but the model has no way to tell them apart because the RUL is the same⁵. This non identifiability means a learned model cannot uniquely map early-life sensor patterns to a specific high RUL. Tree based regressors, for instance, end up averaging over those ambiguous points, while neural networks may overfit to any small spurious differences. Overall, the dataset’s labelling scheme and inherently similar early trajectories produce a one-to-many mapping problem that any model will struggle with in high RUL regions.

3 Imbalanced RUL Distribution and Model Bias

The distribution of training examples across RUL values is highly skewed, which biases models against the high end of the range. In FD001, each engine provides many more data points near failure (Low RUL) than at the very beginning (High RUL), and the capping of RUL exacerbates this imbalance. There are relatively few distinct examples of RUL much above the 125 mark, and virtually non above 130 in the supervised labels³. Models trained on such data tend to focus on the mid to low RUL range (where the target is varying and well represented) and pay less attention to the sparse high RUL cases. For instance, a simple regression model fit on the piecewise target function was observed to be systemically underestimating high RUL:

driven by data towards late predictions (i.e., many predictions in the late part)⁵

This happens because the model sees far more examples of engines in decline and thus leans towards predicting shorter remaining life. Recent works on turbofan RUL have recognized this imbalance issue and introduced remedies. One Transformer-based approach added an adaptive re-weighting of the loss to:

dynamically adjust sample weights of different RUL categories... allowing the model to focus more on difficult-to-learn samples.⁶

By up-weighting the high RUL category, they explicitly counteracted the data imbalance. The need for such techniques underscores that the poor generalisation at high RUL is fundamentally a data distribution problem. Indeed, no matter if one uses RandomForestRegressor, LightGBM, or a deep CNN, a model trained on heavily imbalanced data without correction will yield biased predictions, often erring on the side of lower RUL, when faced with engines early in their life.

4 Summary of Findings

All the above factors illustrate that the high value RUL prediction challenge on C-MAPSS FD001 is rooted in the nature of the dataset itself. Early in life, engines show no clear degradation and share essentially the same label, which confounds learning. The training data also emphasize lower RUL ranges, biasing models toward those regions unless additional handling is applied. These issues occur regardless of model choice: Random Forest and gradient boosting models tend to predict near the maximum label (or the mean) due to lack of early-life signal, while deep learning models often overfit minor sensor noise or simply delay forecasting decline until more obvious trends appear. Researchers have addressed this through data centric approaches such as improved labeling to identify when degradation truly begins, data augmentation to balance the RUL distribution, and engineered temporal features to provide trend context. The consistent finding in the literature is that the problem is not that one model type is inherently weaker, but that all models struggle under the same data constraints. When those constraints are alleviated, models ranging from CNNs to LightGBM perform significantly better at estimating RUL far from failure. In summary, the underperformance and instability of predictions above roughly 200 cycles arises from the scarcity of

early degradation signal and ambiguous early-life labels in FD001, rather than any core limitation of the algorithms themselves. Reliable long-horizon RUL estimation requires providing models with meaningful early-trend information or informed priors for extrapolation.

References

- [1] Prathmesh Pethkar. “RUL Estimation of N-CMAPSS Turbofan Engines using Deep Learning with Customized Penalty and Expanded Sensors”. Master’s Project. San José State University, 2025. DOI: 10.31979/etd.zgmu-6aka. URL: https://scholarworks.sjsu.edu/etd_projects/1523/.
- [2] Kıymet Ensarioğlu, Tülin İnkaya, and Erdal Emel. “Remaining Useful Life Estimation of Turbofan Engines with Deep Learning Using Change-Point Detection Based Labeling and Feature Engineering”. In: *Applied Sciences* 13.21 (2023), p. 11893. DOI: 10.3390/app132111893. URL: <https://www.mdpi.com/2076-3417/13/21/11893>.
- [3] Tilman Krokotsch. *CMAPSS Dataset Reader (RUL Datasets library)*. https://krokotsch.eu/rul-datasets/api/rul_datasets/reader/cmapss/. Accessed: 2025-10-26. 2025.
- [4] Samiha M. Elsherif et al. “A deep learning-based prognostic approach for predicting turbofan engine degradation and remaining useful life”. In: *Scientific Reports* 15.1 (2025), p. 26251. DOI: 10.1038/s41598-025-09155-z. URL: <https://www.nature.com/articles/s41598-025-09155-z>.
- [5] Tarek Berghout and Mohamed Benbouzid. “A Systematic Guide for Predicting Remaining Useful Life with Machine Learning”. In: *Electronics* 11.7 (2022), p. 1125. DOI: 10.3390/electronics11071125. URL: <https://www.mdpi.com/2079-9292/11/7/1125>.
- [6] Zhi Zhang et al. “Multi-Head Self-Attention-Based Fully Convolutional Network for Turbofan Engine Remaining Useful Life Prediction”. In: *Algorithms* 17.8 (2024), p. 321. DOI: 10.3390/a17080321. URL: <https://www.mdpi.com/1999-4893/17/8/321>.